



ELECT MAINT Department
SIMHADRI

Note No :1



Ref:SMTTP/ELECT MAINT/2023-24/EMLMIVFD/442783

Date:31/08/2023(12:52:40)

Subject:LMI on Guidelines on Load Commutated Inverter (LCI) Drive

Subject:New LMI on Guidelines on Load Commutated Inverter (LCI) Drive maintenance

OGN is issued on Guidelines on Load Commutated Inverter (LCI) Drive Maintenance vide Document No [OGN-OPS-ELEC-037](#)

Typographic error on the revision 0 rectified.

Tune SRCLSC is replaced with SRCLFC

According to the above OGN a LMI is prepared for Simhadri project Approving authority is the head of the project

Put up for approval please.

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LMI found in line with OGN 37 rev 1, Dated August 2023.

May Approve further

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Note No:4

reviewed and modified.

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Note No:5

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Note No:8

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Note No:10

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ELECT MAINT- OFF SITE Department
SIMHADRI

Subject:New LMI on Guidelines on Load Commutated Inverter (LCI) Drive
maintenance

Sub:- LMI on Guidelines on Load Commutated Inverter (LCI) Drive Maintenance.

OGN is issued on Guidelines on Load Commutated Inverter (LCI) Drive Maintenance vide Document No
OGN-OPS-ELEC-037

Typographic error on the previous revision rectified. Tune SRCLSC is replaced with SRCLFC
Annexure 1,2 and 3 were modified to checklist format.

According to the above OGN a LMI is prepared for Simhadri project Approving authority is the head of
the project

Put up for approval please.

NTPC Limited

**SIMHADRI SUPER THERMAL POWER
PROJECT**

LOCATION MANAGEMENT INSTRUCTION

LMI/ OGN/OPS/ELEC/037

Rev.1, Aug 2023

**Guidelines on Load Commutated
Inverter (LCI) Drive Maintenance**

APPROVED FOR

IMPLEMENTATION BY - - - - -

HEAD OF THE PROJECT (SIMHADRI)

Date: _____

LMI APPROVAL DOCUMENT

1		Title of LMI	Guidelines on Load Commutated Inverter (LCI) Drive Maintenance
2		a) Document Code b) Revision/Issue No. & Date	LMI/ OGN/OPS/ELEC/037 Rev.1, Aug 2023
3		Original/Ref. document	COS-ISO-00- OGN/OPS/ELEC/037 Aug 2023
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10	Reviewed by	D P Patra	Design : GM- (O & M)
11	Approved for issue	Approved for issue	Design : GGM (Simhadri)

**Guidelines on
Load Commutated Inverter (LCI) Drive Maintenance
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Guidelines on Load Commutated Inverter (LCI) Maintenance

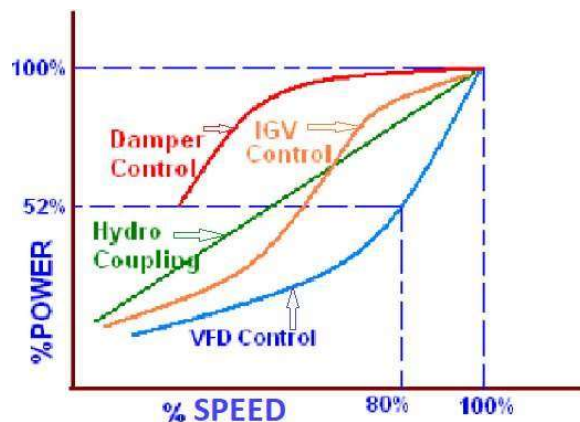
1.0 INTRODUCTION

The Load Commutated Inverter (LCI) is a static and Variable Frequency Drive (VFD) system that controls a synchronous machine from near to rated speed (20% to 100 %). This drive system adjusts fan/pump speed as per process control input signal to get the desired output. In NTPC, this drive system is installed for controlling speed of Induced Draft (ID) fan motors of 250/500 MW BHEL supplied boilers.

LCI drive offers following major benefits over conventional Inlet Guide Vane (IGV) control for ID fan application:

A. Higher Efficiency:

For variable torque loads (like centrifugal pumps, fans and blowers) motor Power (P) $\propto$ speed (N)³, at 75%-80% speed (BMCR condition for ID fan application) power requirement is only around 50%. At 50% fan speed, power requirement is around 12.5%. A small reduction of speed saves significant amount of power consumption. This drive has major contribution in reduction of plant auxiliary power consumption. Power consumption patterns for different controls are shown below.



B. Soft start feature:

LCI drive does not allow motor starting current beyond the rated current. Motor life is enhanced due to lower thermal and mechanical stresses in absence of starting current. There is no limitation in number of starts of motor.

C. Redundancy:

Each ID fan motor has dual stator windings, which are fed from separate LCI channels. Each LCI channel is sized to cater full load requirement of motor in BMCR condition. Hence, each ID fan motor has 100 % redundant LCI drive.

At present, 192 numbers of LCI drive channels are in service in different NTPC and its JV stations. Sometimes the spurious outages of the channels leading to partial loading of units which offsets the above advantages of having LCI drives in ID fan motors. In order to minimize outages of LCI channels, regular monitoring, preventive and annual maintenance plans are formulated in this document.

2.0 SUPERSEDED DOCUMENTS - Nil

3.0 SAFETY INSTRUCTIONS

- 3.1** LCI drive panels are energized from 2.3 KV, 440 V, 230 V AC and 220 V DC supply sources. Only qualified persons having through knowledge of the drive schemes shall access the panels with valid work permit.
- 3.2** As two LCI channels feed different windings mounted on same motor stator, even with one channel running, back EMF of motor may be extended up to other VFD channel. Hence, proper isolations (racking out of source and load breakers) from source and load side of any channel shall be ensured before opening of bridge and blower/filter panels.
- 3.3** The brushless exciter of the motor may be fed either of the channels. The exciter power supplies (440 V AC) of both channels are looped through changeover contactors in the rear side Control and Excitation (C&E) panel of both channels. Hence, proper precaution shall be taken during preventive maintenance and troubleshooting work with other channel / exciter supply is ON.
- 3.4** Before starting of work, proper isolation shall be ensured using non-contact type voltage detector.

4.0 PERIODIC MONITORING AND MAINTENANCE PLAN

Based on the LCI drive fault compendium shared by BHEL and NTPC experience over the years, periodic monitoring and maintenance plans are prepared to improve reliability of the drive. The records of preventive maintenance shall be maintained for review.

Type of inspection	Frequency
Walk down inspection	Daily as per in Annexure-1
Preventive maintenance	Quarterly/Opportunity as per Annexure-2
Overhauling maintenance	Overhauling per Annexure-3

5.0 RECOMMENDATIONS FOR RELIABILITY IMPROVEMENT

- 5.1** A time delay of two seconds shall be given in DDC logic for tripping of FD fan from sensing both channels OFF of corresponding ID fan motor. However, in case of both channel FAULT sensed, the delay for FD fan trip shall be bypassed.
- 5.2** 240 V AC supply for each channel shall be taken separately from UPS distribution board.
- 5.3** Loss of both cooling blower's protection time delay (SQTMR2) shall be increased to 2.0 second from 1.0 second.
- 5.4** Pre-trip NC contact (which changes status during DC control supply failure) in source breaker module shall be shorted.
- 5.5** Replacement of power cables of 20 years of old.
- 5.6** Existing IOMA (old model) modules shall be replaced (phase-wise) with new design input output modules (IOMB and IOPC).
- 5.7** SEM earthing & cable screen earthing is to be followed in line volume-1A recommendation. SEM rack earthing, electronic ZERO Bus, cable screen bus and panel earth buses shall be interconnected and extended to directly earth pit riser from each channel.
- 5.8** Check CB1, CB2 & CB3 thermal timer setting as per manual.

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- 5.9 **SRCLSC** parameter may be increased up to 1.05 from 1.00 to facilitate loading of 3 channels for full load of the unit when one channel is in tripped condition.
- 5.10 Load breaker O/C CAG37 relay setting shall be set higher (1.8PU) than the SUPP1(1.6PU) setting of Load & source side.
- 5.11 All other analogue protection settings shall be in line with sheet 1DC in volume-3 and site commissioning test report.
- 5.12 Separate register/file shall be maintained for each channel for keeping history of events, malfunction, component replacement, preventive maintenance date etc.

6.0 SPARE PARTS MANAGEMENT

- 6.1 Testing instrument per stage: High accuracy true RMS digital multi-meter (02 no), Digital storage oscilloscope, laptop with preloaded software and connector (02 no.).
- 6.2 Following minimum spares are recommended for every unit (four LCI channels):
 One complete set of electronic cards (with programmed EEPROMS), IOMA card, and power supply modules.
- 6.3 Following minimum spares are recommended per two / three units (08/12 LCI channels):
 One number SEM rack (without card). One complete set thyristor arms (Twelve number) and associated electronic modules, one number blower fan, Field exciter module, one number each type CTs, One set of capacitors and MOVs, One number each type of auxiliary contactors and one set of ribbon cables with connectors. One set of Power contactors.
- 6.4 Following minimum spares are recommended for one station:
 One number each type of VFD transformers, one number of each rating reactor, one complete run of power cable with end termination kits required for one motor (two LCI channels), one number load breaker and associated PTs, complete set rotating diode and fuse of exciter, one number each rating ID fan motor.

7.0 RESPONSIBILITY CENTER FOR IMPLEMENTATION

Respective area engineers are responsible for implementation and overall responsibility lies with sectional heads.

Type of inspection	Frequency	Responsibility
Walk down inspection	Daily as per in Anexure-1	Area engineer
Preventive maintenance	Quarterly/Opportunity shutdown as per Anexure-2	Area engineer
Overhauling maintenance	Overhauling per Anexure-3	Area engineer

8.0 RECORD KEEPING

Records will be maintained by concerned area maintenance engineers for walkdown, preventive and overhauling maintenance check lists.

9.0 REVIEW

This LMI document will be reviewed once in two years by Head of the project, NTPC Simhadri.

**COS-ISO-00-OGN/OPS/ELEC037, REV. NO. : 0**

Guidelines on Load Commutated Inverter (LCI) Drive Maintenance

UNIT :-
1/2/3/4

ANNEXURE - 1

DAILY WALK DOWN CHECKLIST

DATE :

Sl. No.	Walk Down Check Points	Status			
		ACH 1	ACH 2	BCH 1	BCH 2
01.	Conduction status of thyristor in source and load converter arms in bridge panel by viewing indication lamp from outside.				
02.	Any VFD alarm is persisting.				
03.	LED status of HISA, HRDD, HROA, IMOK in SEM rack, EXCITER Module.				
04.	VFD room temperature is being maintained between 20°C				
05.	Recording of currents of all channels in load breaker panel ammeter. If any current hunting is observed.				
06.	VFD room false ceiling, windows and doors are in sealed condition.				
07.	Reactor temperatures are to be recorded.				
08.	Motor winding temperatures are to be recorded.				
09.	Reactor room cleanliness and ventilation system working condition.				
10.	Reactor doors are sealed.				
11.	Cable trench condition (no waterlogging condition should persist)				
12.	VFD transformer conservator oil level and breather desiccant condition.				
13.	VFD transformer marshalling box and field protection device terminal box proper sealing.				
14.	Any kind of chattering / electrical discharge sound near transformer LV side terminal box / Reactor / CLC panel (reactor room).				
15.	Cleanliness of VFD room floor.				

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ANNEXURE-2

QUARTERLY/OPPORTUNITY PREVENTIVE MAINTENANCE PLAN

Precautions/safety measures:

Sl. No.	OPPORTUNITY checkpoints	Remark
01.	Exciter power supply (415VAC) of slave channel shall be ensured. It should be verified by measuring with multimeter behind C&E panel.	
02.	No alarm should persist	
03.	Source and load breakers shall be made OFF and racked out. Exciter supply of intended channel shall also be made OFF and module to be locked.	
04.	Switch OFF control supply in PSFF module.	
05.	Remove (and put insulation tape on wires) following wires for the channel taken for maintenance job	
06.	IOMATB5- 09,10: Other channel running signal (Sht1BD/2BD grid 61)	
07.	IOMATB7-10,11: Current reference to other channel (Sht1BF/2BF grid 55)	
08.	IOMATB7-16,17: Current reference into this channel (Sht1BF/2BF grid 67)	
09.	IOMATB2-15,16 This channel running (PTDPS (Sht1BC/2BC grid 61))	
11.	IOMATB1-5, Motor above minimum speed (Sht1BG/2BG grid 17)	
10.	Speed demand signal from DDC shall be looped (for Ch-1 IOMA TB7- 8,9 to be shorted Ref sht 1BF grid 53. For Ch-2, IOMA TB7- 8,9 to be shorted Ref sht 2BF grid 53)	
12.	Excitation supply, UPS supply, blower supply MCBs shall be made OFF	
13.	Common control panel shall be in locked condition while other channel in service.	
14.	As the exciter supply is looped in (downstream of field contactor) both channel, maintenance work in power supply terminals (rear side of C&E panel) shall be avoided while other channel in service.	

Maintenance Activities:

Commutated Inverter (LCI) Drive Maintenance**Bridge panel:**

Sl. No.	OPPORTUNITY checkpoints	Remark
01.	Cleaning of bridge panel front mesh.	
02.	Spare cable glands are sealed.	
03.	Cleaning and inspection of bridge and fan & filter panels in front and rear side. While cleaning with hand blower, the air flow should be regulated so that jumpers, connectors and lugs are not loosened.	
04.	Source and load side CT secondary connections are to be tightened	
05.	Gate supply card(HPTK), voltage attenuation card(XPTN) and cell monitoring card(NTDA) cards are to be cleaned with dry soft brush. Cards shall be inspected for tightness of connectors / push type lugs and sign of component heating/dry soldering/other abnormalities	
06.	Individual cell resistances (between Anode- cathode and Gate - cathode) of source and load side thyristor cells shall be recorded and compared with earlier values. If any cell is found shorted/ very low resistance (in order of ohm range in anode-cathode or shorted Gate-cathode) in a converter branch, the complete arm has to be replaced. Replacement of faulty thyristor in an arm at site is discouraged. However, in case of urgent requirement, replacement of faulty thyristor shall be done as per OEM procedure.	
07.	Tightness checking of DC bus and ground bus loose connections on push type lugs/connectors.	
08.	Visual inspection of MOVs, CTs and support insulators.	
09.	Closing and locking of Bridge panel doors from front & rear side.	

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Fan and Filter panel:

Sl. No.	OPPORTUNITY checkpoints	Remark
01.	Regulated air blowing of wire wound resistors, capacitors and MOVs.	
02.	Inspection of capacitors for oil leakages or any other abnormalities.	
03.	Inspection of wire wound resistors for sign of heating/integrity of soldering joints.	
04.	Closing and locking off and filter panel doors from front & rear side.	

Reactor:

Sl. No.	OPPORTUNITY checkpoints	Remark
01.	Reactor cleaning shall be done by air blowing.	
02.	Sealing reactor doors.	

Control and Excitation (C&E)panel:

Sl. No.	OPPORTUNITY checkpoints	Remark
01.	All electronic cards in SEM rack shall be cleaned soft brush (air blower usage for card cleaning is not recommended).	
02.	Spare cable glands are sealed.	
03.	Cards shall be inspected for any sign component heating/deterioration	
04.	Card locking bar shall be fixed firmly. Locking of SEM rack electronic cards shall be ensured by hand feeling. If required, separate packing may be used to prevent card vibration.	
05.	SEM rack support hinge shall be checked tight fastening with the panel structure.	
06.	Locking and nomenclature of ribbon cable connectors (SEM rack, IOMA, PSFF, XTFH, XTFY cards) shall be verified to ensure proper and correct cables are connected to specific cards.	
07.	Tightness of cable at different control cable terminals of IOMA card, C&E panel TBs and auxiliary contactors shall be checked.	
08.	SEM rack cards are properly locked.	

Normalization Checks:

Sl. No.	OPPORTUNITY checkpoints	Remark
01.	Normalization of control power supply sources including exciter power supply source and power up electronic cards.	
02.	Checking of regulated control voltages (P5, P15, N15 and P28) in SEM rack test pins. Zero voltages at AC OM & DC OM test pins. It is recommended to keep P28V around 30VDC so that it does not come below 26 V while firing pulses are ON.	
03.	Checking of DC offset value of at NLIB (IC –source, 1D-load) test point TP-4, TP-5&TP-6.DC Offset values shall be adjusted (by R1, R2& R3 pots) to 0.0mv.	
04.	Verify the output of the PS7A gating power supply module is 84 V DC ± 5 V. If output is not matching, ADJUST POT 'R1' to get the exact value. Gating supply (without gate firing) = 84 VDC and Gating supply (with gate firing) ≈ 80 VDC.	
05.	Verify the output of the P105 IOMA power supply is 105 VDC $\pm 10\%$.	
06.	Check voltages in test pins on HRIA cards (IOMA slot no 2A, 2B).The voltages shall be either 0V or 30VDC. Any deviations (say between 5V-25VDC) indicate stracking in IOMA motherboard. This checking is applicable to old version IOMA module.	
07.	Carry out Self –test of Electronic cards	
08.	Ensure locking of bridge and fan &filter panel doors.	
09.	Calibration of load side voltages and DC off-set by back charging (by closing of load breaker) while other cannel in service.	
10.	Calibration of source side voltages and DC off-set by closing source breaker.	

Signature

ANNEXURE-3**OVERHAULING MAINTENANCE PLAN**

Precautions/safety measures

Sl. No.	OH check points	Remark
01.	Fan shut down shall be taken	
02.	Source and load breakers shall be made OFF and racked out. Exciter supplies of both channels shall also be made OFF and modules are to be locked.	
03.	Excitation supply (for both channels), UPS supply, blower supply MCBs shall be made OFF.	

OVERHAULING MAINTENANCE PLAN

Maintenance Activities

Bridge panel:

OVERHAULING MAINTENANCE PLAN

Sl. No.	OH check points	Remark
01.	Cleaning and inspection of bridge and fan & filter panels in front and rear side. While cleaning with hand blower, the air flow should be regulated so that jumpers, connectors and lugs are not loosened.	
02.	Tightness checking of power cable and bus-bar junction points shall be done.	
03.	Visual inspection of MOVs, CTs and support insulators. IR checking of insulation.	
04.	Source, load side and isolation CT secondary connections are to be tightened.	
05.	Gate supply card (HPTK), voltage attenuation card (XPTN) and cell monitoring card (NTDA) cards are to be cleaned with dry soft brush. Cards shall be inspected for tightness of connectors / push type lugs and sign of component heating/ dry soldering /other abnormalities.	
06.	Individual cell resistances (between Anode- cathode and Gate -cathode) of source and load side thyristors cells shall be recorded and compared with earlier values. If any cell is found shorted/ very low resistance (in order of ohm range in anode-cathode or shorted Gate-cathode) in a converter branch, the complete arm has to be replaced. Replacement of faulty thyristors in an arm at site is discouraged. However, in case of urgent requirement, replacement of faulty thyristors shall be done as per OEM procedure.	
07.	Checking of Insulation Resistance of Thyristor Bus.	
08.	Insulation Resistance and AC high pot test (up to rated voltage of cable) checking of all power cables (between Transformer to LCI, LCI to reactor, LCI to Load contactor and load contactor to motor). Cables with IR value less than 200 MΩ are to be replaced. Leakage currents of all cables for each section shall be almost equal. Cables with higher leakage current shall be replaced.	
09.	Tightness checking of DC bus and ground bus loose connections on push type lugs/connectors.	
10.	Single point grounding of power cable screens in different panels (Bridge, reactor, CLC and load contactor) shall be ensured. This may be verified by checking insulation resistance of the cable screens w.r.t ground after removing it from grounding strip.	
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Commutated Inverter (LCI) Drive Maintenance

Maintenance Activities

Fan and Filter panel:

Sl. No.	OH check points	Remark
01.	Regulated air blowing of wire wound resistors, capacitors and MOVs.	
02.	Inspection of capacitors for oil leakages or any other abnormalities.	
03.	Inspection of wire wound resistors for sign of heating/integrity of soldering joints.	
04.	Preventive maintenance of blower fan motor and vibration checking shall be carried out.	

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Commutated Inverter (LCI) Drive Maintenance
OVERHAULING MAINTENANCE PLAN

Maintenance Activities

Control and Excitation (C&E) panel:

Sl. No.	OH check points	Remark
01.	All electronic cards in SEM rack and SFF module (Exciter Controller) shall be cleaned soft brush (air blower usage for card cleaning is not recommended).	
02.	Cards shall be inspected for any sign component heating/deterioration.	
03.	Check all parameters and Jumper setting as per manual.	
04.	Card locking bar shall be fixed firmly. Locking of SEM rack electronic cards shall be ensured by hand feeling. If required, separate packing may be used to prevent card vibration.	
05.	SEM rack support hinge shall be checked tight fastening with the panel structure.	
06.	Locking and nomenclature of ribbon cable connectors (SEM rack, IOMA, PSFF, XTFH, XTFY cards) shall be verified to ensure proper and correct cables are connected to specific cards.	
07.	Dry soldering sign shall be verified in all burden resistors connections of different electronic cards (NLCB, DLCB, XTFH, NSFC, DLIB etc).	
08.	Tightness of cable at different control cable terminals of IOMA card, C&E panel TBs and auxiliary contactors shall be checked.	
09.	Cleaning, inspection and tightness checking of exciter controller card, thyristors and connectors.	
10.	Checking of healthiness of Field Exciter module & SEM rack cooling fans.	
11.	Cleaning and inspection of IOMA card back plane shall be done.	
12.	DCS interface cable continuity and logic shall be checked.	
13.	Earth pit resistance shall be checked. Integrity of grounding strip from LCI panel to earth-pit, intra-panel grounding and inter panel earth bus connection shall be checked.	

OVERHAULING MAINTENANCE PLAN

Maintenance Activities

Common control panel:

Sl. No.	OH check points	Remark
01.	Cleaning and tightness checking of panel TBs and terminals of auxiliary contactors.	
02.	Checking of cable gland sealing.	
03.	Insulation resistance checking transformer field protection alarm and trip cables.	
04.	DCS interface cable continuity and logic shall be checked.	

Load contactor/breaker:

Sl. No.	OH check points	Remark
01.	Cleaning and tightness checking of panel TBs, terminals of auxiliary contactors and breaker auxiliary contacts.	
02.	Preventive maintenance of Load breaker.	

Reactor panel:

Sl. No.	OH check points	Remark
01.	Cleaning and inspection of reactor coil.	
02.	Ensure Sealing of reactor panel.	
03.	Insulation Resistance checking of reactor.	

OVERHAULING MAINTENANCE PLAN

Maintenance Activities

Motor Exciter inspection:

Sl. No.	OH check points	Remark
01.	Motor exciter shall be cleaned and inspected for looseness, sign of overheating and any other abnormalities.	
02.	Insulation Resistance of the exciter cable from LCI panel shall be checked. Phase sequence of the exciter cable shall be marked before isolation.	
03.	Rotating diodes reverse blocking characteristics shall be checked	
04.	Torque tightness of DC ring bus connectors.	

VFD Transformer Maintenance:

Sl. No.	OH check points	Remark
01.	Check for transformer oil leakages and oil level.	
02.	Transformer oil moisture level, DGA and BDV shall be checked.	
03.	IR checking LV to E, HV-LV, HV to E.	
04.	Transformer field protection terminal box seal is to be done.	
05.	Sealing of Transformer marshalling box shall be ensured.	

OVERHAULING MAINTENANCE PLAN

Normalization Checks

Sl. No.	OH check points	Remark
01.	Normalization of control power supply sources including exciter power supply source and power up electronic cards.	
02.	Auxiliary supply changeover to be checked.	
03.	Checking of regulated control voltages (P5,P15,N15 and P28) in SEM rack test pins. Zero voltages at ACOM & DCOM test pins. It is recommended to keep P28 Varound 30V DC so that it does not come below 26 V while firing pulses are ON.	
04.	Checking of DC offset value of at NLIB (IC –source, 1D-load) test point TP-4, TP- 5 & TP-6. DC Offset values shall be adjusted (by R1, R2 & R3 pots) to 0.0 mv.	
05.	Verify the output of the PS7A gating power supply module is 84 V DC $\pm$ 5 V. If output is not matching, ADJUST POT 'R1' to get the exact value. Gating supply (without gate firing) = 84 VDC and Gating supply (with gate firing) = $\approx$ 80 VDC	
06.	Verify the output of the P105 IOMA power supply is 105 V DC $\pm$ 10%.	
07.	Check voltages in in test pins on HRIA cards (IOMA slot no 2A, 2B). The voltages shall be either 0V or 30 VDC. Any deviations (say between 5V- 25 V DC) indicates tracking in IOMA mother board.	
08.	Carry out Self-test of electronic cards.	
09.	Carry out GATE TEST.	
10.	Source & load side flux calibration with source supply.	
11.	Carry out crow bar test.	
12.	Excitation test.	
13.	EVC power test.	

LMI/OGN/OPS/ELEC/037, Rev. No.: 1 Aug 2023 Simhadri
Commutated Inverter (LCI) Drive Maintenance

EVC Power Test Table:

Channel-1

Sl. No.	Exciter voltage (volts)			Exciter current (Amps)			IFB-(2A) Vdc
	R-Y	Y-B	B-R	R	Y	B	
1							
2							
3							
4							
5							
6							
7							
8							

Channel-2

Sl. No.	Exciter voltage (volts)			Exciter current (Amps)			IFB-(2A) Vdc
	R-Y	Y-B	B-R	R	Y	B	
1							
2							
3							
4							
5							
6							
7							
8							

Signature

Commissioning Procedure

- (a) Normalization of control power supply sources including exciter power supply source and power up electronic cards.
- (b) Auxiliary supply changeover to be checked.
- (c) Checking of regulated control voltages (P5,P15,N15 and P28) in SEM rack test pins. Zero voltages at ACOM & DCOM test pins. It is recommended to keep P28 V around 30V DC so that it does not come below 26 V while firing pulses are ON.
- (d) Checking of DC offset value of at NLIB (IC –source, 1D-load) test point TP-4, TP-5 & TP-6. DC Offset values shall be adjusted (by R1, R2 & R3 pots) to 0.0 mv.
- (e) Verify the output of the PS7A gating power supply module is 84 V DC ± 5 V. If output is not matching, ADJUST POT 'R1' to get the exact value. Gating supply (without gate firing) = 84 VDC and Gating supply (with gate firing) = ≈ 80 VDC
- (f) Verify the output of the P105 IOMA power supply is 105 V DC $\pm 10\%$.
- (g) Check voltages in in test pins on HRIA cards (IOMA slot no 2A, 2B). The voltages shall be either 0V or 30 VDC. Any deviations (say between 5V- 25 V DC) indicates tracking in IOMA mother board.
- (h) Carry out Self-test of Electronic cards.

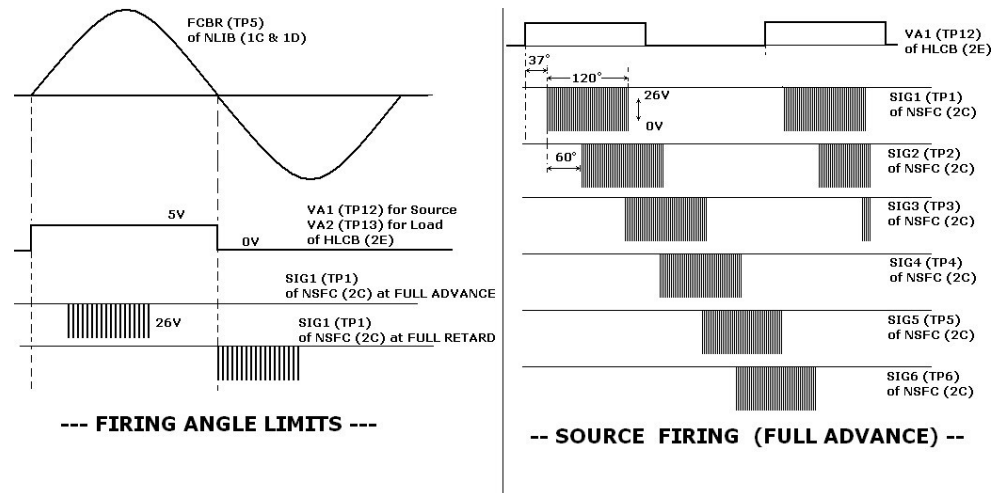
(i) Carry out GATE TEST:

The gate test mode allows OFF-LINE testing of the analog signal conditioning, microprocessor firing control and gate pulse generation circuitry. Thyristor gate pulses may be verified for proper magnitude and phasing.

METHOD:

- ✓ Cooling blower shall be made ON.
- ✓ Temporarily change values of **MTRFMIN to 0.2 & FLUXS to 0.1**. It should not be saved.
- ✓ Put the NLCB TEST Switch in UP position. Check the RPM meter on the panel deflects and shows a reading. Verify **GTSTM is true "T"**.
- ✓ Wait for about 5 minutes.

- ✓ All Gate pulse LEDs in all the thyristor power modules should glow. If they do not, then the DC offset voltages in FAB, FBC & FCA in NLIB source and LOAD is excessive.
- ✓ Use BAL1 & BAL2 pots in NLCB to make the zero DC offsets as minimum as possible. value of less than 50mV DC should suffice.
- ✓ Check P28 voltage shall be more than 26 V DC while firing pulses are ON.
- ✓ Using a current probe check the gate current waveform is OK. This can be done by comparison between waveforms with other thyristors.



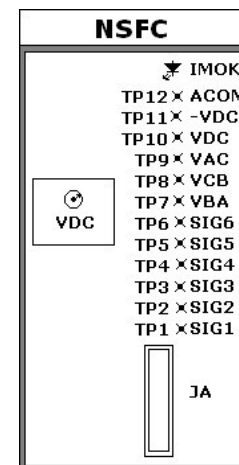
- ✓ Put NLCB switch in down “**DOWN**” position. Gate pulses should turn off. Switch off panel blowers. Change **MTRFMIN & FLUXS** to tune-up values.
- ✓ Reset microprocessor by temporarily shorting RST and DCOM on HMPG. This ensures that the old vales of **MTRFMIN and FLUXS** have been restored.

(j) Source & load side flux calibration with source supply:

- ✓ Verify all the doors are closed.
- ✓ Verify that SWGR relay on the I/O module will trip the incoming switchgear in the event a fault ‘FT1’ is detected by the control (may be checked by Switching OFF the Blowers.

Source voltage balancing:

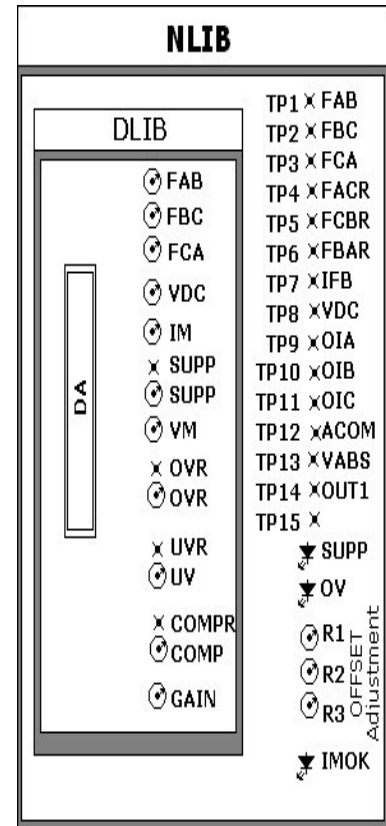
- Close source side circuit breaker. Observe that source side bridge neon lamps are illuminated.
- Measure following values on the NSFC (2C) card.
- Measure AC Voltage measured at TP7 (VBA)
- Measure AC Voltage measured at TP8 (VCB)
- Measure AC Voltage measured at TP9 (VAC)
- The actual system voltage = (ATTN RATIO) X VAC
- (ATTN RATIO is given on the system elementary sheet 1EA)
- FBAR = (Actual system voltage /nominal system voltage) X 4.75 V+ ac(rms)
- + Value shall be cross checked with OEM manual Vol-3 sheet-1DC.



- Measure voltage at TP4 (FBAR) on the NLIB-IC
- (Adjusted by R1 pot to obtain voltage as calculated above)
- Measure voltage at TP5 (FCBR) on the NLIB-IC
- (Adjusted by R2 pot to obtain voltage as calculated above)
- Measure voltage at TP6 (FACR) on the NLIB-IC
- (Adjusted by R3 pot to obtain voltage as calculated above)
- (** FBAR, FCBR & FACR signals should be balanced as accurately as possible (+/-5mV))
- Measure DC Offset voltage at TP4 on NLIB-1C
- Measure DC Offset voltage at TP5 on NLIB-1C
- Measure DC Offset voltage at TP6 on NLIB-1C
- (** DC offset voltages should be less than 100 mV)
- Measured voltage (-dc) at TP13 (VABS) of NLIB-1C
- Measured voltage (ac) at TP9 (VAC) of NSFC-2C =
- ** Verify that VABS (-dc) level is approximately 2.229 X VAC at TP9 of NSFC-2C

Load side signal phasing and balance:

- ✓ Trip source circuit interrupter.
- ✓ Disconnect ribbon cable JA from NSFC- 2B.
- ✓ Move the ribbon cable JA from NSFC- 2C to NSFC- 2B. Connect a test jumper from backplane pins 1C- 55 to 1C- 1(if necessary) to disable source under voltage trip.
- ✓ Close source circuit breaker. Observe that source side bridge neon lamps are illuminated.
- Measure following values on the NSFC (2B) card.
- Measure AC Voltage measured at TP7 (VBA)
- Measure AC Voltage measured at TP8 (VCB)
- Measure AC Voltage measured at TP9 (VAC)
- The actual system voltage = (ATTN RATIO) X VAC
(ATTN RATIO is given on the system elementary sheet 1EA)
- FBAR = (Actual system voltage /nominal system voltage) 4.75 V~ ac(rms)
~Value shall be cross checked with OEM manual Vol-3 sheet-1DC.
- Measure voltage at TP4 (FBAR) on the NLIB-1D
- (Adjusted by R1 pot to obtain voltage as as given in sheet 1DC for load flux function)
- Measure voltage at TP5 (FCBR) on the NLIB-1D
(Adjusted by R2 pot to obtain voltage as given in sheet 1DC for load flux function)
- Measure voltage at TP6 (FACR) on the NLIB-1D
- (Adjusted by R3 pot to obtain voltage as as given in sheet 1DC for load flux function)
- (** FBAR, FCBR & FACR signals should be balanced as accurately as possible (+/-5mV))
- Measure DC Offset voltage at TP4 on NLIB-1D



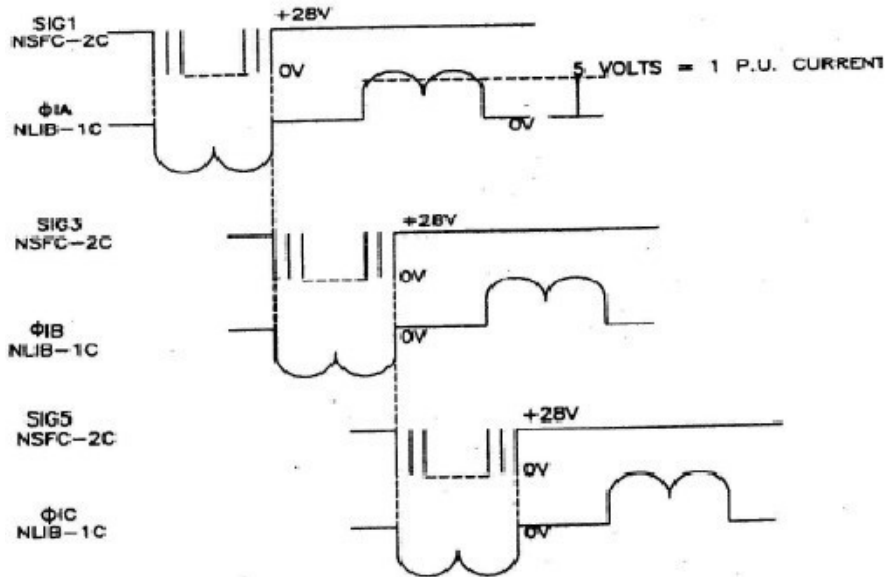
- Measure DC Offset voltage at TP5 on NLIB-1D
- Measure DC Offset voltage at TP6 on NLIB-1D
- All protecting settings shall be verified with respect to the OEM manual Vol-3 sht no 1DC.
(* DC offset voltages should be less than 100 mV)
- Measured voltage (-dc) at TP13 (VABS) of NLIB-1D =
- Measured voltage (ac) at TP9 (VAC) of NSFC-2B =
** Verify that VABS (-dc) level is approximately 2.229 X VAC at TP9 of NSFC-2B
** It is necessary to fine tune FBAR, FCBR, FACR when running the motor on load. Check motor volts with respect to speed on flux profile, to ascertain this value.

(k) Carry out crow bar test:

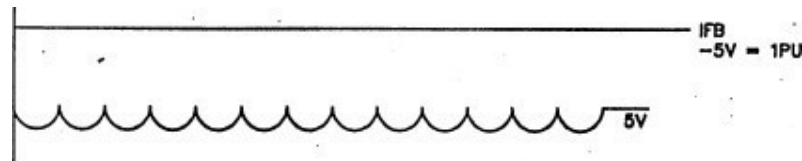
The crowbar test permits controllable DC link current to be circulated without involving the motor windings. On the load side the DC current circulated through one or more diametric cell pairs by passing the motor.

The signals to be observed on the oscilloscope will indicate, whether source side CTs are phased correctly. If the CTs are not phased correctly, reverse the CT lead connections at the CT terminals or any other convenient point in the CT feedback Circuit. Note that in crowbar test, there will not be any current though the load side CTs. In a successful crowbar test, there should not be any deflection on the motor voltmeter on the panel.

- ✓ Apply source power to the bridge. Clear all faults.
- ✓ Load beaker is isolated.
- ✓ Connect a terminal to the HMPG RS 232 port.
- ✓ Modify the following EEPROM variables.
- ✓ CRMIN to 0
- ✓ SQDRTST to 68
- ✓ CRSTART to 0
- ✓ Connect Ch-1 of the oscilloscope to NSFC-2C, SIG1 (TP1)
- ✓ Connect Ch-2 of the oscilloscope to NLIB-1C, OIA (TP1)
(similarly check for SIG3 & OIB and SIG5 & OIC)
- ✓ Initiate a RUN sequence.
- ✓ Modify CRSTART to 0.1 Pu.
- ✓ Check the current feedback signal IFB-1C (TP7) =
 - CRSTART=0.1
 - IFB=-0.5Vdc + or -0.2 Vdc
- ✓ Increase CRSTART to 0.2 pu. Current should be continuous.
- ✓ Observe the test point pairs for all three current and gating signals as indicated in Fig6.1 (VFD O&M manual vol.-1A) (also shown below).



TYPICAL ANALOG WAVEFORM (Current Signals & Gate Pulse Signals)



TYPICAL CURRENT FEEDBACK SIGNAL ('IFB' TEST POINT) ON NLIB

Note: To carry out crowbar test with other channel in service following additional modifications to be done and restored after test.

- ✓ IOMA TB5- 09,10: Other channel running signal (Sht 1BD/2BD grid 61)
- ✓ IOMA TB7- 10,11: Current reference to other channel (Sht 1BF/2BF grid 55)
- ✓ IOMA TB7- 16,17: Current reference into this channel (Sht 1BF/2BF grid 67)
- ✓ IOMA TB2- 15,16 , This channel running/RTPOS (Sht 1BG/2BG grid 61)
- ✓ Ensure Load side Breaker/Contactor does not close by switching OFF supply in Breaker/Contactor. Short MISOL input at IOMA TB4-05 ,06 (Sht 1BD/2BD grid 21). Remove this short after Crowbar test.
- ✓ Modify tune-ups SQDRTST, CRSTART, CRMIN back to their original value.

(I) Excitation test:

These tests verify the operation of the exciter voltage controller (EVC). Verify the burden resistors on the NHVL board in the EVC are as per the elementary sheet 1EA.

Phase sequence check:

- ✓ Apply power to the source bridge and to the EVC input terminal.
- ✓ Connect an oscilloscope to observe simultaneously the waveforms on the test points ATTEN- A1 (TP1), ATTEN- B1 (TP3), ATTEN- C1 (TP5) on the NHVL board.
- ✓ Confirm the phase sequence is 1-2-3 and that all three signals are balanced.
- ✓ With power on the source bridge, use an oscilloscope to compare source VAC (TP9-2C) against EVCL- VAC (TP9-2A).
- ✓ Measure the number of degrees VAC (TP9-2C) leads or lags VAC (TP9-2A). Record this value and save into tune up setting FANGLAG.
FANGLAG = ** Allowable value for FANGLAG range from -45 to +45 degrees

(m) EVC power test:

- ✓ Apply source bridge power and EVC input power.
- ✓ Attach a voltmeter across one phase of the exciter output.
- ✓ Attach a clamp on ammeter around one phase of the Exciter output.
- ✓ Select LOCAL operation and set the local reference pot to minimum.
- ✓ Open the Flux regulator and select EVC test mode by setting tune up SQFEC to 2.
- ✓ The EVC firing is now controlled by local speed pot.
- ✓ Using the local speed pot (IOMA) increase the exciter current to max allowable exciter current feedback at test point IFB-2A read as follows.
IFB = Measured EVC current / rated EVC current * 5 Vdc
IFB = _____
Adjusted at test point IFB-2A is equal to _____ V dc.
- ✓ Using clamp on meter, verify that all three output currents and voltages are equal.

Sl. No.	Exciter voltage (volts)			Exciter current (Amps)			IFB-(2A) Vdc
	R-Y	Y-B	B-R	R	Y	B	
1							
2							
3							
4							
5							
6							
7							
8							

- ✓ Reduce exciter current to zero and modify SQFEC back original value.
- ✓ Normalize flux regulator.